

HORNSUND, Svalbard and Jan Mayen

WMO: 010030

Lat: 77.000N

Lon: 15.500E

Elev: 39

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-7.8	-4.9	-18.2	2.0	-6.6	-14.6	2.5	-3.7	42.4	17.8	38.5	18.7	10.7	80	0.996

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	46.6	42.9	45.0	42.0	43.9	41.3	43.9	45.5	42.8	44.3	41.8	43.4	15.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
42.8	40.6	44.1	41.6	38.7	43.1	40.5	37.1	42.3	17.1	45.8	16.5	44.4	16.1	43.5	50.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
37.1	33.0	29.4	-9.4	50.4	5.4	1.8	-13.3	51.7	-16.5	52.7	-19.5	53.7	-23.5	55.0		
			-10.3	46.4	5.2	1.6	-14.1	47.5	-17.1	48.5	-20.0	49.4	-23.8	50.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	18.3	17.8	13.4	19.2	28.6	36.7	40.7	39.9	35.9	28.9	24.8	20.1	(6)
(7)		DBStd	11.89	11.54	10.35	10.44	9.37	5.38	2.78	2.27	2.48	4.33	6.71	7.84	9.75	(7)
(8)		HDD50	8371	984	901	1136	923	663	400	288	314	423	655	756	928	(8)
(9)		HDD65	13846	1449	1321	1601	1373	1128	850	753	779	873	1120	1206	1393	(9)
(10)		CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	12.6	14.9	16.0	14.1	13.4	12.4	9.2	10.0	9.6	11.8	11.4	14.3	14.7	(14)
(15)	Precipitation	PrecAvg	15.5	1.5	1.2	1.2	0.8	0.7	0.7	1.1	1.5	2.0	1.6	1.7	1.5	(15)
(16)		PrecMax	29.0	3.8	3.2	2.3	2.5	1.9	2.8	3.6	3.4	5.6	5.6	6.6	4.5	(16)
(17)		PrecMin	9.4	0.3	0.2	0.2	0.2	0.1	0.0	0.0	0.3	0.6	0.4	0.5	0.4	(17)
(18)		PrecStd	3.7	0.9	0.8	0.6	0.6	0.4	0.6	0.9	0.9	1.3	1.1	1.3	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	37.5	35.7	37.6	38.7	45.2	50.3	48.8	45.6	42.9	39.9	38.2	(19)
(20)			MCWB	36.0	35.9	34.0	35.6	36.0	40.7	45.2	44.2	43.5	41.6	38.0	36.4	(20)
(21)		2%	DB	35.6	35.1	34.0	35.3	36.9	43.2	47.1	46.5	44.2	41.4	37.6	35.9	(21)
(22)			MCWB	34.2	33.5	32.5	33.8	34.5	39.6	42.9	43.2	42.4	40.1	35.5	34.3	(22)
(23)		5%	DB	33.9	33.6	32.2	33.7	35.8	41.8	45.4	44.9	43.0	39.9	36.1	34.5	(23)
(24)			MCWB	32.6	32.0	30.9	32.0	33.4	38.8	42.0	42.1	41.3	38.6	34.1	32.8	(24)
(25)		10%	DB	32.3	31.7	29.2	31.9	34.8	40.5	44.1	43.6	41.9	38.2	34.7	33.0	(25)
(26)			MCWB	30.8	30.0	27.5	30.4	32.5	37.9	41.3	41.1	40.2	36.8	32.8	31.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	36.2	34.4	36.0	37.2	42.3	45.7	45.8	44.4	42.2	38.6	36.8	(27)
(28)			MCDB	37.2	37.2	35.2	36.8	38.2	44.0	49.2	47.6	45.0	42.5	39.4	37.6	(28)
(29)		2%	WB	34.5	34.1	32.8	34.0	35.3	40.5	43.8	43.9	43.0	40.4	35.9	34.7	(29)
(30)			MCDB	35.3	35.0	33.9	35.2	36.4	42.2	45.9	45.7	43.7	41.0	37.3	35.7	(30)
(31)		5%	WB	32.9	32.1	31.0	32.2	34.0	39.3	42.7	42.6	41.7	38.7	34.4	33.1	(31)
(32)			MCDB	33.8	33.5	32.3	33.5	35.3	41.1	44.5	44.3	42.7	39.8	35.8	34.3	(32)
(33)		10%	WB	31.0	30.0	27.6	30.5	32.9	38.3	41.8	41.4	40.4	36.9	33.0	31.6	(33)
(34)			MCDB	32.2	31.6	29.2	31.9	34.4	40.2	43.6	43.1	41.7	38.1	34.6	32.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	7.6	8.5	7.5	5.1	4.5	5.3	5.2	4.9	5.8	6.7	7.7	(35)
(36)			MCDBR	7.0	7.2	9.9	7.0	5.5	6.5	7.5	7.1	5.4	5.9	6.1	7.1	(36)
(37)			MCWBR	7.1	6.9	9.9	6.5	4.5	4.6	5.4	5.2	5.0	5.8	5.8	7.0	(37)
(38)		5% WB	MCDBR	7.6	7.4	10.0	7.8	4.8	5.9	6.8	6.7	5.6	6.0	6.3	7.7	(38)
(39)			MCWBR	7.9	7.3	10.0	7.7	4.5	4.5	5.4	5.4	5.3	6.1	6.1	7.8	(39)
(40)		Clear-Sky Solar Irradiance	taub	N/A	0.224	0.275	0.297	0.312	0.302	0.306	0.297	0.261	0.367	N/A	N/A	(40)
(41)	taud		N/A	1.888	2.045	2.195	2.255	2.378	2.424	2.425	2.360	2.722	N/A	N/A	(41)	
(42)	Ebn at Noon		0	20	173	241	260	272	265	246	199	37	0	0	(42)	
(43)	Edh at Noon		0	3	18	27	30	28	26	22	14	4	0	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0	11	225	745	1215	1501	1190	797	301	42	0	0	(44)	
(45)		RadStd	0	2	31	108	94	134	115	83	36	5	0	0	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (3 neighbors)		+2.74	+8.33	+7.38	N/A	N/A	N/A	-994	-988	N/A	N/A	

Nomenclature: See separate page